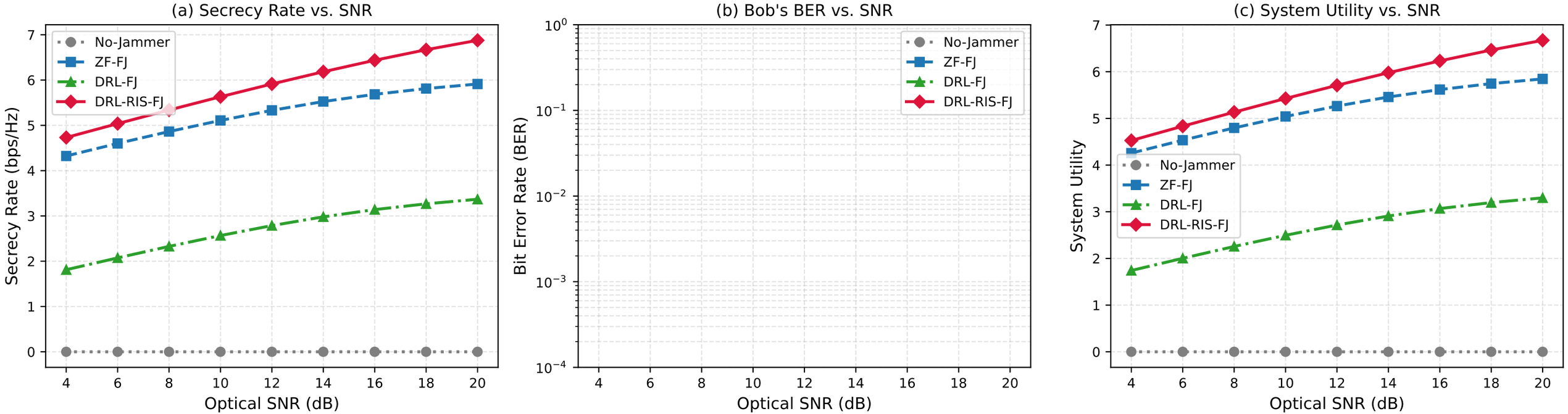Performance Evaluation across Transmit Optical SNR Range (4 dB to 20 dB)